APOFormer: Adaptive Geometric Primitive Optimization Transformer for Building Polygon Extraction From Remote Sensing Imagery

Qinglie Yuan

***Abstract*—The automatic generation of building polygons from remote sensing imagery has significant applications but confronts challenges due to complex urban scenes, frequent occlusions, multiscale features, and diverse geometric morphologies. Segmentation-based methods are constrained by irregular boundaries and imprecise contours, often requiring extensive postprocessing. Geometric primitive-based approaches typically overlook geometrical primitive interdependencies and contextual semantic correlations, causing suboptimal accuracy and topological errors. To address these issues, this study proposed a novel approach, Adaptive Geometric Primitive Optimization Transformer (APOFormer), for building polygon extraction from remote sensing imagery, combining the advantages of the convolution neural network (CNN) and Transformer. APOFormer represents polygons as vertex-query-based sequences and establishes interactive mechanisms for primitive correlations using positional embedding and image context. In feature extraction, the CNN encoder incorporated learnable corner-edge detection operators to explicitly enhance building primitive features. Additionally, a dynamic vertex query mechanism and the multi-vertex collaborative deformable attention (DA) module are developed to improve position prediction accuracy and efficiency using the Transformer-based decoder. Furthermore, an edge-context-guided topology reconstruction (ECTR) module is constructed to enhance polygon quality. Comprehensive experimental evaluation on WHU, INRIA, and MASB datasets, validated by visualization and qualitative analyses, demonstrates the superiority of APOFormer, achieving an average of 83.79% Average Precision (AP), 27.73 MTA, 88.01% C-IoU, and 93.90% IoU. APOFormer achieves remarkable generalization ability and provides robust solutions for accurate building polygon generation.**

***Index Terms*—Building polygon, deep learning, geometrical primitive, remote sensing, transformer.**

I. Introduction

THE automatic interpretation of building vector polygons from high-resolution remote sensing imagery has broad applications in cartography, autonomous navigation, urban planning, and cadastral surveying [1], [2], [3], [4], [5]. Building contours are formally represented as vertex sets with explicit topological relationships, requiring algorithms to infer the spatial positions, semantic categories, and geometric dependencies of each building instance [6], [7]. Nevertheless, the complexity of urban scenes, coupled with the heterogeneous spectral features and diverse building morphologies, remains a significant challenge for accurate and generalizable building polygon extraction frameworks [8], [9].

Existing mainstream approaches contain two categories: segmentation-based and geometric primitive-based methods. The former employs heuristic pipelines that use semantic segmentation to generate binary building masks, which are often followed by post-processing steps such as vectorization, regularization, or contour simplification. Deep neural networks have enhanced the ability to extract segmentation features [10]. However, semantic segmentation only provides pixelwise footprints without offering explicit instance differentiation. To address this limitation, several studies have adopted instance segmentation frameworks, such as SOLOV2 [11], Mask R-CNN [12], CenterMask [13], and PolarMask [14] for instance-level object detection and polygon extraction.

The above methods predominantly depend on segmentation performance and often suffer from topological inconsistencies and shape distortions in the vectorization. Fine-grained building features could become blurred or poorly reconstructed (e.g., rounded corners and irregular edges). To mitigate these issues, several studies have incorporated geometric constraints or energy-based formulations into ground-truth labels and model training processes to refine feature learning [15], [16]. However, these approaches are not end-to-end and still rely on postprocessing and manual adjustments, causing low efficiency and redundant feature computation.

The latter often decomposes buildings into geometric primitives, such as corners and lines, and predicts their topological structure and categories. For instance, Acuna et al. [17] and Zhao et al. [18] employ implicit propagation to capture the sequence dependencies between polygon vertices, ensuring that the generated contours have topological closure and geometric coherence. Compared with segmentation-based approaches, these paradigms generally yield more geometrically regular and topologically consistent shapes. However, these methods encounter challenges in training stability and positional optimization due to their low computational efficiency and cumulative error propagation across multiple iterative steps.

Alternatively, hybrid frameworks integrating the superiority of different deep network architectures have been developed.

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PolyWorld [19] uses convolution neural network (CNN) to generate candidate vertices and visual descriptors, with an attention-based graph neural network (GNN) inferring topology via adjacency permutation matrices. Jiao et al. [20] proposed an R-CNN framework predicting vertex sequences and extracting instance-level features via RoIAlign. Wang et al. [21] utilized the segment anything model (SAM) with RoIAlign to generate polygons, predicting masks, Gaussian vertices, and boundaries.

Recently, Transformer-based architectures have been adopted in semantic segmentation [22], [23], using multihead self-attention (MHSA) to capture long-range dependencies. Improved MHSA algorithms (such as DETR [24], deformable DETR [25], DABDETR [26], and Efficient DETR [27]) use coordinate sequences as query embeddings for object detection. These end-to-end, query-based models eliminate anchors and nonmaximum suppression, improving efficiency. Applied to building polygon prediction, they directly infer vertex coordinates and topology without postprocessing. For instance, PolyBuilding [28] uses an encoder–decoder architecture and corner query refinement. RoIPoly [29] adopts query-based extraction, constraining vertex-query attention within RoIs. Line2Poly [30] applies line primitives for coarse-to-fine topology reconstruction. P2PFormer [31], [32] integrates multiscale RoI features with multiprimitive query groups.

Geometric primitive-based methods offer high inference efficiency and accuracy via end-to-end paradigms, but some issues remain.

1) The existing studies directly extract building geometric features without an optimization strategy, introducing background noise and category ambiguity in decoders.
2) Current approaches often focus on a single vertex context, but do not model the interdependencies within each building instance.
3) Building primitive decoders that employ fixed query embeddings, failing to accommodate diverse building shapes, and redundant queries compromise network efficiency.
4) Moreover, existing methods only consider vertex features but ignore boundary context presentation for topology reconstruction.

To address these issues, this study proposed an Adaptive Geometric Primitive Optimization Transformer (APOFormer) for building polygon extraction. Compared with existing methods, APOFormer explicitly integrates learnable corner-edge geometric primitives and cross-level feature fusion, suppressing background noise and category ambiguity. Meanwhile, the encoder dynamically optimizes vertex queries via confidence priors instead of fixed query embeddings, adapting to diverse building shapes and accelerating convergence. Decoder models the interdependency among vertices within the instance via multi-vertex collaborative deformable attention (DA), overcoming the constraints of single-vertex context modeling in prior works. Furthermore, polygon topology is reconstructed by edge context rather than relying solely on vertex features, effectively improving topological consistency. These integrated designs form a pure end-to-end framework without postprocessing. The model code is available at https://github.com/yuanqinglie/building-polygon-extraction.git. The main contributions of this article are as follows.

1) An end-to-end CNN–Transformer integrated architecture is constructed for building polygon extraction. CNN-encoder performs multiscale feature extraction and geometric query embedding, while Transformer-decoder handles contextual interaction and topological optimization, enabling polygon prediction from imagery without any postprocessing.
2) An end-to-end CNN–Transformer integrated architecture is constructed. CNN module performs multiscale feature extraction and geometric query embedding, while the Transformer conducts contextual query interaction and topological optimization. This enables direct prediction of building polygons from remote sensing images without any postprocessing.
3) To address the limitations in random query initialization, a dynamic query optimization mechanism is introduced based on CNN encoder priors. Candidate vertex queries are selected via a confidence-based strategy to balance accuracy and convergence efficiency.
4) A multivertex collaborative DA mechanism is developed to simultaneously aggregate multiple positional features and context. Additionally, edge context-guided topological reconstruction can effectively improve predictive performance.

II. Methodology

A. Network Architecture

Fig. 1 depicts the overall network architecture employing an encoder–decoder paradigm. The encoder using the CNN-based structure takes remote sensing imageries as inputs to generate multiscale feature representations as L^i ($i \in \{2,3,4,5\}$), each downsampled by a 2^i resolution to inputs. Meanwhile, candidate vertices as query initialization are predicted via primitive optimization. The decoder constructs positional embedding and iteratively refines semantic features via four stacked DA layers. In a parallel decoding branch, the polygonal topology is reconstructed by aggregating edge features and predicting the corresponding adjacency matrices. Ultimately, the model produces vertex confidence scores, polygon coordinates, and an edge-adjacency score matrix, collectively forming the final polygonal representation.

B. Building Primitive Features Enhancement

Corners and edges are fundamental geometric primitives of buildings, offering robust clues for polygonal vector generation. Most methods use implicit paradigms to extract them, raising decoding complexity in intricate remote sensing scenes. Superfluous background information hinders MHSA optimization, especially in shallow features. To address these issues, this study developed the Crossing-Feature Primitives Enhancement (CFPE) module by incorporating learnable explicit operators and modeling cross-level contextual correlations to optimize edge-corner features.

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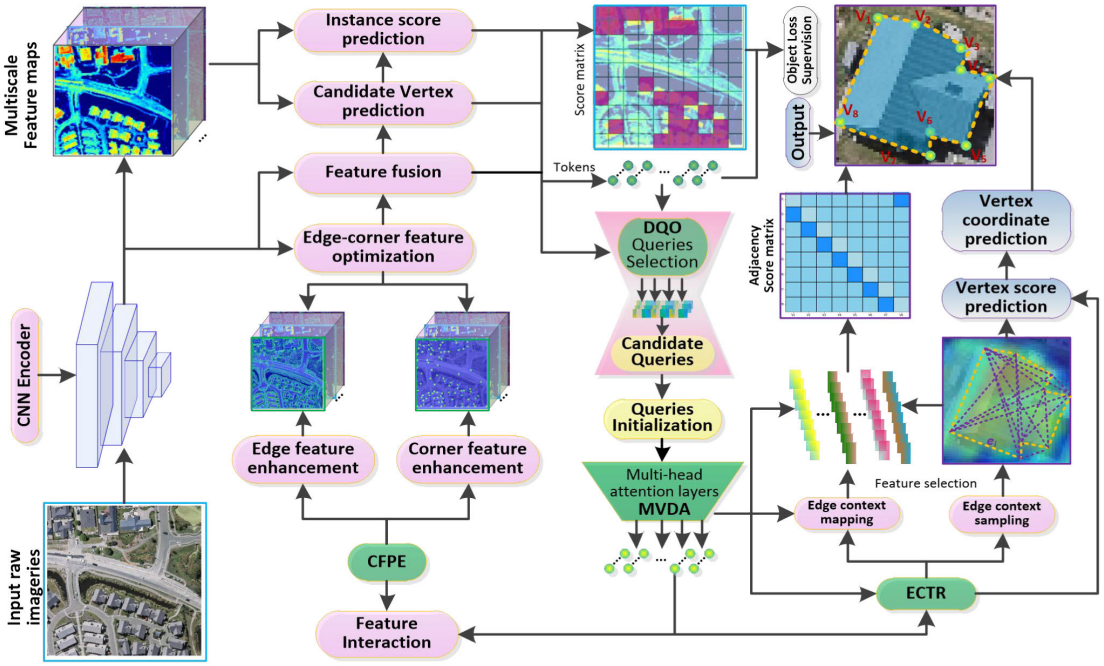


Fig. 1. Overall network pipeline.

As illustrated in Fig. 2, the hierarchical features $Li \in \mathbb{R}^{H_i \times W_i \times C_i}$ are obtained from encoders (i is the index of the encoding stage; H, W, and C are height, width, and channels for feature maps, respectively). To construct multiscale information, the CFPE fuses and maps L_2–L_3 as shallow features $X^l \in \mathbb{R}^{H' \times W' \times C_l}$ and L_4–L_5 as deep features $X^h \in \mathbb{R}^{H' \times W' \times C_h}$, via sampling, addition, and convolution 1×1 ($\mathrm{Conv}_{1\times1}$), as defined by (1) and (2). Additionally, X^l and X^h are aggregated into two channel embedding vectors via average pooling operator $AvgPool(\cdot)$ and multi-layer perceptron (MLP). An affinity matrix $A_{ff} \in \mathbb{R}^{C_l \times C_h}$ is then constructed by embedding vectors using matrix multiplication and the *Softmax* function, as presented in (3). Shallow features can be refined by multiplying by A_{ff}. This operation aims to establish an interactive mechanism between deep semantics and shallow information. A_{ff} describes the affinity of semantic correlation for X^l and X^h. Ultimately, the refined shallow features are fused with high-level features X^h by residual connection, as formulated in (4)

$$X^l = Conv_{1\times1}\left(Add\left(Dows_{\downarrow\times2}\left(L_2\right), L_3\right)\right) \tag{1}$$

$$X^h = Conv_{1\times1}\left(Add\left(Ups_{\uparrow\times2}\left(L_5\right), L_4\right)\right) \tag{2}$$

$$A_{ff} = Softmax\left(MLP\left(AvgPool\left(X^l\right)\right) \otimes MLP\left(AvgPool\left(X^h\right)^{\mathrm{T}}\right)\right) \tag{3}$$

$$Z\left(X^l, X^h\right) = Add\left(Conv_{1\times1}\left(X^l \otimes A_{ff}\right), Ups_{\uparrow\times2}\left(X^h\right)\right) \tag{4}$$

where $Dows_{\downarrow\times2}$ and $Ups_{\uparrow\times2}$ denote two times downsampling and upsampling operation, respectively.

Furthermore, Harris and Sobel algorithms [33] are incorporated into CNN layers to explicitly extract corner and edge features. Instead of fixed template parameters, the CFPE embeds these operators into trainable convolutional layers, enabling the network to adaptively learn optimal detection

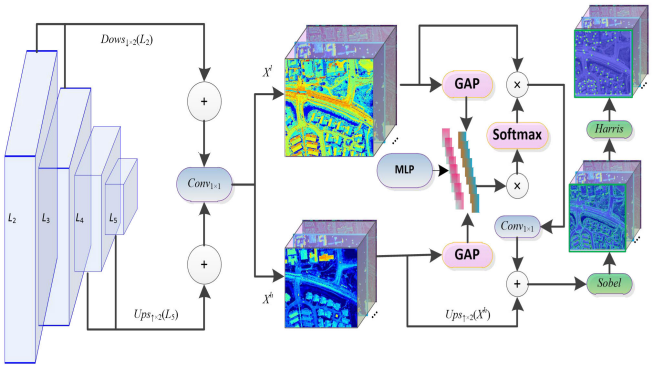


Fig. 2. CFPE.

parameters. Specifically, the trainable convolutional kernels for each feature map are initialized with Sobel filter weights. Depthwise separable convolution (DSC) is adopted to reduce computational complexity. The input is the refined feature map $Z(X^l, X^h)$, and the output is the corner feature map F_{Ha} and the Sobel features F_{so}. The algorithm description is as follows.

DSC operators G_x and G_y are applied to compute horizontal and vertical feature gradients, respectively, as shown in (5). Gradient statistics and a Gaussian smoothing operator, $Gauss(\cdot)$, are performed to suppress noise. Subsequently, squared gradient components and the cross product of gradients are calculated, as defined in (6). Based on the processed gradient components, edge and corner features are computed separately. The edge feature map F_{so} is defined as the gradient amplitude, which quantifies the strength of edge responses at each pixel, as shown in (7). The corner feature map F_{Ha} is computed using the Harris function, which measures the corner response by integrating the gradient statistics, as in (8). In model training, G_x and G_y are fine-tuned to adaptively capture

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the contextual features. Ultimately, the enhanced corner-edge feature map Z_f is generated by fusing $Z(X^l, X^h)$, F_{Ha}, and F_{so} according to (9)

$$G_x = DSC_x\left(Z\left(X^l, X^h\right)\right), \quad G_y = DSC_x\left(Z\left(X^l, X^h\right)\right) \tag{5}$$

$$I_x^2 = G_x^2, \quad I_y^2 = G_y^2, \; I_{xy} = Gauss\left(G_x, G_y\right) \tag{6}$$

$$F_{so} = \sqrt{I_x^2 + I_y^2} \tag{7}$$

$$F_{Ha} = Det(\boldsymbol{M}) - k \cdot tr(\boldsymbol{M})^2 \tag{8}$$

$$Z_f = Conv_{1\times1}\left(Z\left(X^l, X^h\right) + F_{Ha} + F_{so}\right) \tag{9}$$

where:

1) *M:* Is the gradient covariance matrix;
2) *Det(**M**):* Is the determinant;
3) *tr(**M**):* Is the trace of M;
4) $k \in (0.04, 0.06)$*:* Is an empirical constant.

C. Dynamic Query Optimizer

Query-based decoders generate vertex embedding but fail to adapt to building geometric diversity, and redundant initial queries could delay model convergence. To tackle these limitations, this study developed the Dynamic Query Optimizer (DQO), which adaptively initializes vertex queries as illustrated in Fig. 1.

Inspired by efficient DETR [20] and SOLO [4], DQO uses prior information from the CNN encoder as initialization for query sequences. Specifically, the encoding features Z_f are fused with positional embeddings P_e, as defined in (10). In an arbitrary training sample, if N_B and N_v designate the maximum number of buildings and vertices per instance, respectively, the total number of vertices is defined as $N_V = N_B \times N_v$. DQO adopts a dual-branch network: the one branch maps Z_f to candidate vertex coordinates $F_c \in \mathbb{R}^{S\times S\times 2N_v}$, where S is the grid scale. The other maps X^h to instance scores $F_s \in \mathbb{R}^{S\times S\times 2}$ using the *Softmax* function. The final candidate queries are selected by (11). N_B, N_v, K, and S are determined in subsequent experiments. To supervise F_c and F_s, some loss functions are designed as described in Section III-C

$$\begin{aligned} F_c &= Conv_{\downarrow3\times3}(Z_f + P_e) \\ F_s &= Softmax\left(Conv_{3\times3}\left(Pool_{max}\left(X^h\right)\right)\right) \end{aligned} \tag{10}$$

$$Q_c, Dr_c = \left\{Select_{top-K}\left(Z_f^i, F_c \middle| F_s^i\right), i = 1, 2, 3, \ldots, K\right\} \tag{11}$$

where:

1) *K:* Is the number of selected queries;
2) Q_c*:* Denotes the query embedding;
3) Dr_c*:* Denotes the candidate query coordinates;
4) $Conv_{\downarrow3\times3}$*:* Denotes convolution 3×3 with downsampling;
5) $Select_{top-K}(\cdot|\cdot)$*:* Denotes generation top-K queries based on predictive scores F_s and candidate points F_c.

D. Multivertices Collaborative DA

DETR-based methods use MHSA and multihead cross-attention (MHCA) to obtain positional context and integrate image features, but are computationally intensive. Although some optimization strategies have been developed, such as DA, they are primarily designed for object detection using bounding box representations. Conversely, building a polygon requires refinement of multiple coordinate sequences. Therefore, this section proposed multi-vertices collaboratively DA (MVDA) by extending deformable reference points, as illustrated in Fig. 3.

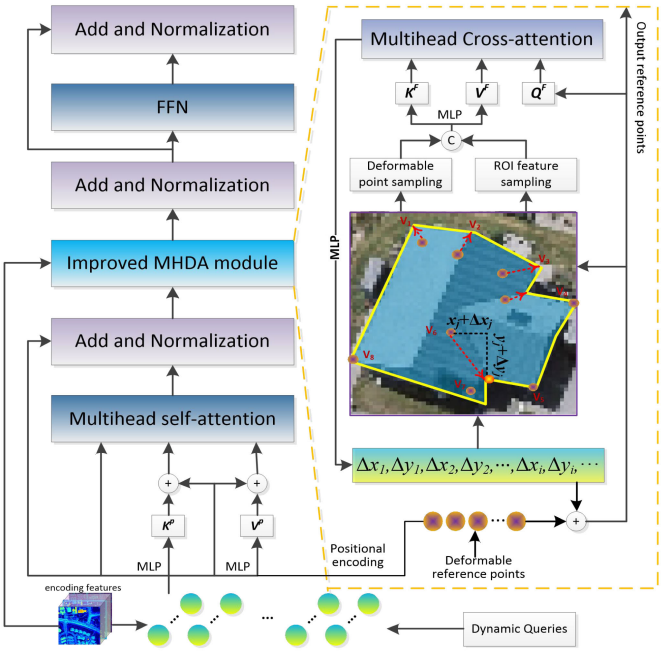


Fig. 3. MVDA.

MVDA consists of MHSA for positional interaction and MHCA for image contextual inference. Q_i is the ith query from Q_c ($i \in \{1, 2, \ldots, N_B\}$). *MLP* maps Q_i to K_i^p, and V_i^p as positional embedding. Hence, MHSA can be applied to each Q_i, as presented in (12), where Q_i^p is positional embedding

$$Q_i^p = Norm\left(Add\left(MHSA\left(Q_i, K_i^P, V_i^P\right), Q_i\right)\right). \tag{12}$$

Generally, DA constructs contextual query features by predicting a compact group of reference points. Unlike DA, this research extends these reference points to each query group Q_i and deformable reference coordinates Dr_c in MVDA and projects their coordinates onto the feature map for sampling. When calculating the attention between vertices queries, MVDA considers all vertices belonging to the same object as a set of reference points. Thus, each vertex not only focuses on its own context, but also on the local features around other vertices.

All vertex queries share sampling features within the same building instance to enhance spatial correlation. Additionally, the maximum and minimum coordinates are obtained from Dr_c to determine each ROI region as an auxiliary context F_i^{roi}, as defined in (13) and (14). F_i^{roi} is sampled as a fixed scale feature map with 7×7 pixel dimension and concatenated with the vertex sampling feature$F_{i,j}^q$(i denotes the index of Q_i and j denotes the index of the vertex). This design integrates building vertices features and ROI region features to enrich contextual representation. Ultimately, each vertex query Q_i' is

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updated via the attention operator $MHCA(\cdot)$ using F_i^{roi} and $F_{i,j}^{q}$, as denoted in (15)–(17). The decoder stacks six MVDA blocks to output the vertex queries. Finally, each vertex reference is updated as Dr_c', as shown in (18) and (19)

$$F_{i,j}^{q} = BIS\left(Z_f | Dr_c\right), \quad j \in 1,2,\ldots,N_v, \ i \in 1,2,\ldots,N_B \tag{13}$$

$$F_i^{roi} = Conv_{3\times 3} \times \left(Pool_{max}\left(BIS\left(X^h | \Omega^{roi}_{\max\left(\left\{Dr_{ci}^{j}\right\}\right),\min\left(\left\{Dr_{ci}^{j}\right\}\right)}\right)\right)\right) \tag{14}$$

$$F^{qr} = Concat\left(\left\{F_{i,j}^{q}\right\}, \left\{F_i^{roi}\right\}, j \in 1,2,\ldots,N_v, i \in 1,2,\ldots,N_B\right) \tag{15}$$

$$Q^F = MLP\left(Q_i^p\right), \quad K^F, V^F = MLP\left(F^{qr}\right) \tag{16}$$

$$Q_i' = Norm\left(Add\left(MHCA\left(Q^F, K^F, V^F\right), Q_i^p\right)\right) \tag{17}$$

$$\Delta Dr_{(\Delta x_j, \Delta y_j)} = MLP\left(Q_i'\right) \tag{18}$$

$$Dr_c' = Dr_c + \Delta Dr_{(\Delta x_j, \Delta y_j)} \tag{19}$$

where:

1) $\Delta Dr_{(\Delta x_j, \Delta y_j)}$: Denotes coordinate offsets;
2) $BIS(\cdot|\cdot)$: Demotes bilinear interpolation sampling;
3) Ω: Is the ROI coordinate domain;
4) Q^F, K^F, and V^F are query, key, and value in MHCA, respectively;
5) $Concat\ (\cdot)$: Demotes feature concatenation.

E. Edge-Context-Guided Topology Reconstruction

The adjacency matrix is commonly used to predict topological relationships among all vertices. However, such predictions often rely on vertex embeddings while neglecting building boundary context, which harbors rich spatial features to guide edge-connection restoration. Thus, this study proposed an edge-context-guided topology reconstruction (ECTR) module for optimal edge connection prediction, as shown in Fig. 4.

Instead of vertex context embeddings, this ECTR integrates edge sampling features and generates robust vertex-edge embeddings. If a polygon contains a maximum of vertices N_V, it can generate candidate edges N_e (where $N_e = N_V(N_V-1)/2$). Top-e edges E^Q can be selected from Dr_c based on confidence scores S^e, as denoted in (20) and (21). Meanwhile, the edge feature is extracted from the Z_f map. A multiscale pooling operator, $MSP(\cdot)$, is introduced into this operation. $MSP(\cdot)$ performs spatial feature sampling along the edge direction using max pooling to capture sufficient context. For aggregating multiscale boundary information, this study uses multiple pooling kernels with 3, 5, and 7 to extract edge features. Then, these features are concatenated and aggregated

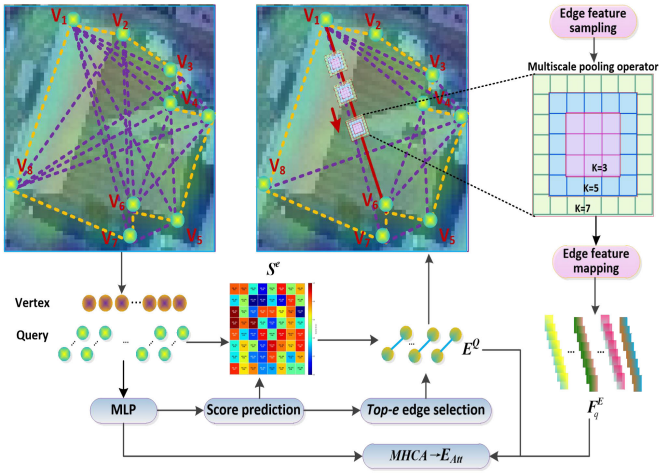


Fig. 4. Edge-context-guided topology reconstruction.

using the addition with normalization. MLP maps them to construct contextual embedding, as presented in (22)

$$S^e = Softmax\left(MLP\left(Concat\left(Q_i, Q_{i+1}\right)\right)\right) \tag{20}$$

$$E^Q = Select\left((Dr_c)|S^e\right) \tag{21}$$

$$F_q^E = Norm\left(Add\left(MLP\left(MSP\left(Z_f | E^Q\right)\right)\right)\right). \tag{22}$$

MHCA is applied to generate an enhanced context for the pairwise vertices. Based on the research in Section II-D, the vertex queries Q can be obtained via MVDA. MLP function maps Q and F_q^E into Q^e, K^e, and V^e, as defined by (23). The attention map E_{Att} can be obtained using (24), where $Q^e \in \mathbb{R}^{N_V \times C}$, $K^e \in \mathbb{R}^{e \times C}$, $V^e \in \mathbb{R}^{e \times C}$. Finally, MLP functions map E_{Att} into two adjacency matrices with clockwise A_{clo} and counterclockwise A_{cou}, respectively. The score matrix S_A can be obtained by the addition of A_{clo} and transposed A_{cou}, as denoted in (25)

$$Q^e, K^e, V^e = MLP\left(Q, F_q^E\right) \tag{23}$$

$$E_{Att} = Norm\left(Add\left(MHCA\left(Q^e, K^e, V^e\right), Q^e\right)\right) \tag{24}$$

$$S_A = A_{clo}\left(MLP_{clo}\left(E_{Att}\right)\right) + A_{cou}\left(MLP_{cou}\left(E_{Att}\right)^{\mathrm{T}}\right). \tag{25}$$

The groundtruth binary adjacency matrix is constructed based on a manually annotated polygon, as illustrated in the left panel of Fig. 5. In directional graphs, rows and columns of the matrix correspond to source and target nodes, respectively. A value 1, denoted by a blue cell, indicates the presence of an edge connection, whereas a value 0 denotes nonadjacency. The clockwise traversal of vertices V_1 through V_9 forms the closed-loop polygon. This matrix serves as the supervision benchmark to quantify the topological consistency between model predictions and groundtruth. On the right panel, each element in permutation matrices A_{clo} or A_{cou} represents the confidence score for an edge connection. Notably, A_{clo} is theoretically identical to the transpose of A_{cou}. For instance, an edge V_5–V_6 corresponds to the clockwise and counterclockwise connection probability scores in the A_{clo} and A_{cou}, respectively. Therefore, two matrices provide a bidirectional probabilistic description for the topological polygon structure.

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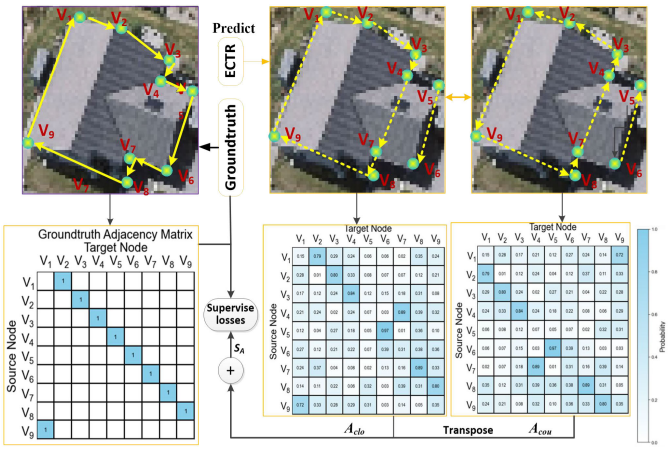


Fig. 5. Polygon topological adjacency matrix construction.

III. Experimental Implementation

The ablation study and comparative experiments were conducted in this section to evaluate the performance of the proposed model. Details of the experimental implementation are described below.

A. Experimental Configuration

All model training was configured with the AdamW optimizer and conducted on the PyTorch platform using an NVIDIA RTX 4060 GPU. The hyperparameters are initialized with a learning rate of 1e-4, batch size of 8, and decay of 1e-3. The learning rate was reduced adaptively if the validation accuracy remained stable for more than 3 epochs. The data augmentation strategy is applied to enhance the generalization performance of the algorithm by randomly rotating, scaling, and mirroring the training datasets.

B. Dataset Description and Evaluation Metrics

The proposed model was comprehensively evaluated on three public building polygonal interpretation datasets, including the WHU_ aerial image dataset (denoted as WHU) [34], Inria Aerial Image Labeling dataset (denoted as INRIA) [35], and Massachusetts building dataset (denoted as MASB) [36]. These datasets have complex remote sensing scenarios, including vegetation cover, diverse land cover types, and shadow changes in urban or suburban areas. Buildings exhibit various geometric shapes and are distributed at different densities. These factors pose challenges to the inference performance of different models.

The WHU dataset consists of aerial remote sensing orthorectified images with red-green-blue (RGB) bands and a high spatial resolution of 7.5 cm, covering 450 km^2 in Christchurch, New Zealand. A total of 229 065 polygonal building instances are provided with accurately delineated contours.

The INRIA dataset primarily comprises 360 aerial orthophotos with a size of 5000 $\times$ 5000 pixels, covering 810 km^2 from different cities in the United States and Austria. These orthophotos are processed using RGB bands with a ground resolution of 0.3 m. A subset of 180 orthorectified images containing over 200 000 buildings was selected.

The MASB dataset includes 151 aerial orthophotography images and more than 300 000 buildings covering 340 km^2 in Boston, United States. Each image has a size of 1500 $\times$ 1500 pixels and a 1-m spatial resolution with RGB bands. Some topological errors and missing building polygons have been corrected with reference to visual interpretation and OpenStreetMap.

To facilitate unified network training, these orthoimages were resampled to a spatial resolution of 0.3 m and cropped into fixed-size tiles of 512 $\times$ 512. Meanwhile, the dataset was split into training, testing, and validation sets at a ratio of 7:2:1. Three types of metrics were used in the experiment to comprehensively evaluate the performance of the proposed method.

1) Intersection over Union (IoU) was applied in experiments to evaluate the accuracy of building footprint extraction. IoU was calculated by comparing the number of pixels contained in the ground truth labels with the predicted polygon coverage.
2) Average precision (AP) is used to evaluate the discriminative ability of building instances. In addition, AP_{50} and AP_{75} were used in the experiment to test the sensitivity of the model detection based on the IoU threshold.
3) Max tangent angle error (MTA) [37] and Complexity Awareness IoU (C-IoU) [19] are used to evaluate the geometric performance of polygons, such as shape regularity and geometric complexity.

MTA can quantify the irregularity of the polygon contour and the deviation of its shape from the ground truth. Specifically, MTA calculates the maximum value of the tangent angle difference between the corresponding edge segments of the predicted polygon and the ground truth polygon. The tangent angle of each edge segment reflects the direction change of the polygon contour. A larger tangent angle difference indicates a more obvious deviation in the contour direction, which directly causes a more irregular polygonal shape. Thus, a higher MTA value implies poorer geometric consistency and lower shape quality.

C-IoU integrates coverage accuracy and vertex complexity. A higher C-IoU score indicates better polygon quality. It inherits the advantage of IoU in measuring the overlap degree between the predicted and ground truth regions, ensuring the accuracy of the building footprint coverage. Additionally, it introduces a vertex complexity that quantifies the rationality of vertex distribution. From a perceptual perspective, a polygon with high C-IoU not only has a high overlap with the real building footprint but also has a regular contour and reasonable vertex distribution, avoiding redundancy or deficiency of vertices.

The ablation experiment used average metrics (including mIoU, mAP, mC-IoU, and mMTA) on three datasets to comprehensively evaluate the performance of different modules. Furthermore, the experiment uses conventional metrics to evaluate the efficiency of the model: Parameter (M) denotes the number of parameters in millions and measures the model size. GFLOPs denotes the Giga floating-point operations per

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second, measuring the model complexity. Inference time (ms) denotes the consumed time for forward inference per image, measuring the inference speed.

C. Model Training Losses

To ensure the model training performance, a multitask loss strategy is designed to integrate constraints from vertex detection, coordinate regression, topology reconstruction, and polygon regularization. The total loss $\mathcal{L}_{total}$ is defined as a weighted sum of sub-losses, as defined in the following equation:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{ved} + \lambda_2 \mathcal{L}_{ver} + \lambda_3 \mathcal{L}_{top} + \lambda_4 \mathcal{L}_{por} \tag{26}$$

where λ_1–λ_4 are balancing coefficients (empirically set as 1.0, 2.0, 1.5, 1.0, respectively, and fine-tuned based on validation performance). $\mathcal{L}_{ved}$ loss supervises the DQO to generate accurate candidate vertices, considering both their classification confidence and spatial localization within the minimum bounding rectangle (MBR) of each building instance, as defined in (27), where $\mathcal{L}_{focal}$ is the focal loss; $s \in [0,1]^{S\times S}$ is predicted confidence scores and $s^* \in \{0,1\}^{S\times S}$ is ground-truth mask; MBR^* is the ground-truth MBR, computed as the maximum and minimum vertex coordinates; K is the number of building instances; $\mathcal{L}_{SmoothL_1}$ is the smooth L1 function

$$\begin{cases} \mathcal{L}_{ved} = \mathcal{L}_{focal}(s, s^*) + \dfrac{1}{K}\displaystyle\sum_{j=1}^{K} \mathcal{L}_{SmoothL_1}\left(MBR_j, MBR_j^*\right) \\ MBR^* = \left[x_{min}^*, y_{min}^*, x_{max}^*, y_{max}^*\right] \end{cases} \tag{27}$$

where $\mathcal{L}_{ver}$ is the vertex regression and classification loss using the L1 and Focal loss, ensuring positional alignment with ground-truth polygons, as defined in (28), where c is the predicted vertex and c^* is its groundtruth; $\hat{p}$ is the predicted vertex coordinates after the MVDA refinement and p_i^* is the ground-truth vertex coordinates; N_v is the number of vertices per building after optimal matching via the Hungarian algorithm to handle order inconsistency. $\mathcal{L}_{top}$ and $\mathcal{L}_{por}$ are the topology reconstruction loss and polygon regularization loss referenced to PolyWorld and PolyBuilding. $\mathcal{L}_{top}$ supervises ECTR module to predict accurate adjacency relationships between vertices, using permutation matrices derived from the score matrix S_A. $\mathcal{L}_{por}$ loss enforces geometric regularity of predicted polygons, including angular constraints, pixel-level alignment, and vertex count

$$\mathcal{L}_{ver} = \frac{1}{N_v}\sum_{i=1}^{N_v} \left\| p_i{}^{\wedge} - p_i^* \right\|_1 + \mathcal{L}_{focal}(c, c^*). \tag{28}$$

IV. Results and Discussion

A. Module Ablation Experiments

1) Encoding Feature Enhancement Performance: Based on the study in Section II-B, the encoders employ CFPE to enhance corner-edge features and provide clues for the decoder. CFPE consists of three parts: cross-feature interaction (CFI), corner feature enhancement (CFE), and edge feature enhancement (EFE). In this experiment, the above components were ablated separately to evaluate the optimization performance of CFPE on encoding features. The backbone network (BN) without CFPE was used as a comparison and provides fused encoding features X_l and X_h as inputs for other modules.

TABLE I
Ablation Accuracy Results Using CFPE on the Datasets

Ablation components		Metrics			
		m-IOU	m-AP	mC-IoU	mMTA
BN+	None	89.87	80.42	85.66	32.69
	CFI	91.21	82.17	85.63	32.57
	CFE+CFI	91.67	82.34	85.78	31.97
	EFE+CFI	93.45	83.22	87.67	28.14
	CFPE	93.90	83.79	88.01	27.73

Table I reports the ablation results using CFPE on the datasets. CFI improved the model's detection and extraction accuracy by 1.34% mIoU and 1.75% mAP, but had no significant impact on the polygonal quality, referring to mMTA and mC-IoU. This confirms that CFI in hierarchical encoders can optimize the transmission of contextual and spatial information. However, the structural features have not been enhanced, especially for topological connections in vertex-edge.

Conversely, EFE+CFI can reduce MTA to 28.14 and increase C-IoU to 87.67%. Although CFE + CFI improved IoU and AP compared to BN, they did not surpass EFE + CFI in terms of polygonal quality. This manifests that EFE has a greater impact on CFPE than CFE. Possibly, CFE constructed corner feature responses via edge gradients sparsely distributed in the spatial feature, which is insufficient to provide richer context than boundary features. Nevertheless, these layers achieved better gains compared to the BN with an increase of 3.37% mAP, 4.03% mIoU, 2.35% mC-IoU, and reduce 4.96 mMAT.

Some feature maps are visualized using the Grad-CAM algorithm on test samples, as shown in Fig. 6. The heat maps are constructed by calculating the average along the channel dimension. The features of edges and corners are significantly enhanced when using CFE and EFE. In pre-CFPE, shallow features contained many spatial details confused with the background. Some redundant background information shows a weak feature response when using CFPE, while the edge-corner areas around buildings show a robust response, as illustrated in the red rectangle. This indicates that the introduction of high-level semantic context via CFPE can guide the refinement of edge-corner features.

2) Optimal Queries Initialization: In the DQO group, parameter K determines the optimal number of queries for initialization. These candidate queries were selected based on the predicted scores using CNN layers. The maximum value of K is set to 100 and $S = 20$. The ablation experiment included five control groups with K values increasing at intervals of 20 to evaluate the performance of dynamic queries. In the initial stage, the queries are randomly initialized, where $K = 0$ indicates that the queries are treated as random training parameters, similar to the DETR configuration.

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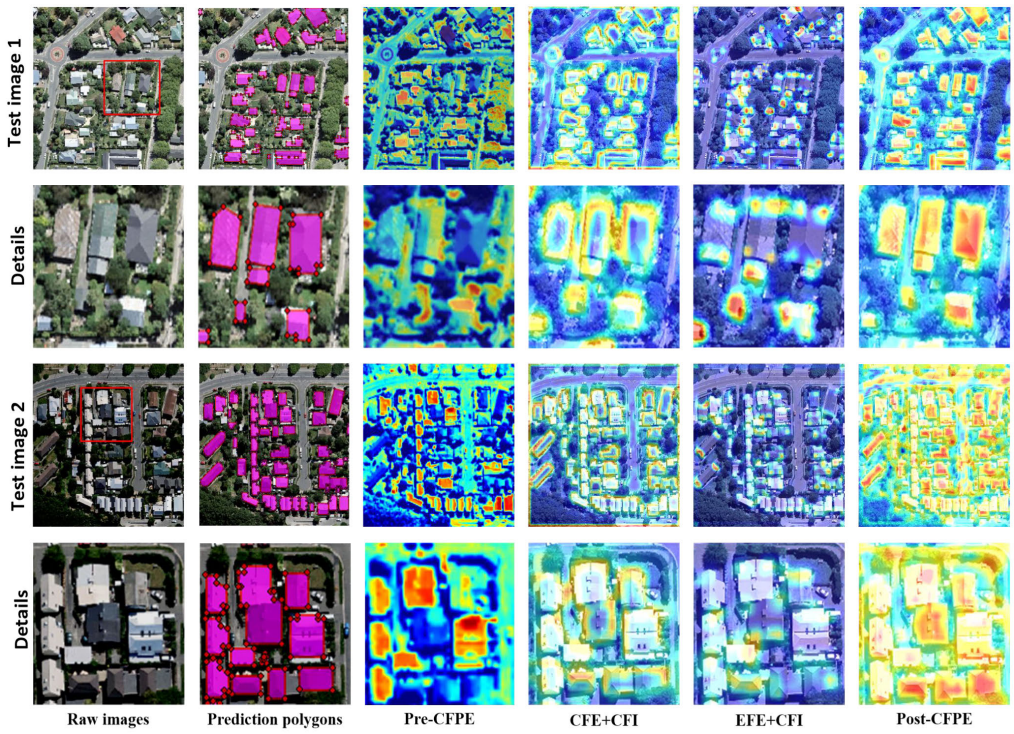


Fig. 6. Visualization of heat map using CFPE. Some details are enlarged in the red rectangle on the second and fourth rows.

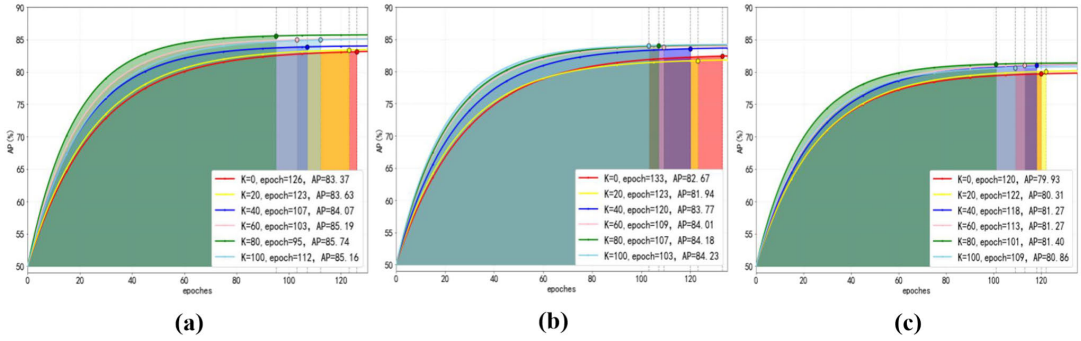


Fig. 7. Model efficiency evaluation using DQO. (a)–(c) AP curve graphs on the WHU, INRIA, and MASB datasets, respectively.

As presented in Fig. 7, no significant variations were observed in accuracy metrics or model convergence when $K < 20$. However, when K exceeded 40, AP exhibited a noticeable increasing trend. Additionally, training iterations gradually decreased as K increased. These results indicate that introducing prior queries can improve prediction efficiency and accuracy. In the initial stage, random query parameters with fewer K fail to capture diverse building objects. As prior queries increase, the decoder can obtain sufficient candidate vertices and accelerate the convergence process.

When K exceeded 80, AP and the convergence efficiency exhibited a slight decline. Probably, the optimal queries (high-confidence ones corresponding to building instances) approached saturation as increasing K yielded no additional informative priors. Additionally, the redundant queries increase the decoder's optimization complexity. Their encoding introduced noise into the query embedding, thereby weakening the supervisory effect for the CNN. Consequently, $K = 80$ was adopted as the optimal setting, considering the trade-off between detection accuracy and convergence efficiency.

3) Deformable Vertex Context Evaluation: The multihead DA (MHDA) mechanism provides deformable reference points and sampling features for each query to reduce algorithm complexity. In contrast, MVDA utilizes multiple query vertices as reference points from the same building instance. These reference points collaboratively update information for the attention mechanism. In feature aggregation, each vertex query simultaneously integrates other vertex features and ROI information. Therefore, this experiment ablates the MVDA module and sets up two control groups, including only using MHDA and ROI features.

As shown in Table II, mIoU and mAP increased by 2.24% and 3.53%, respectively, compared to the BN+ROI when using MVDA. Meanwhile, MAT is decreased by 2.86, while C-IoU is improved by 1.78%. Fig. 8 presents different prediction

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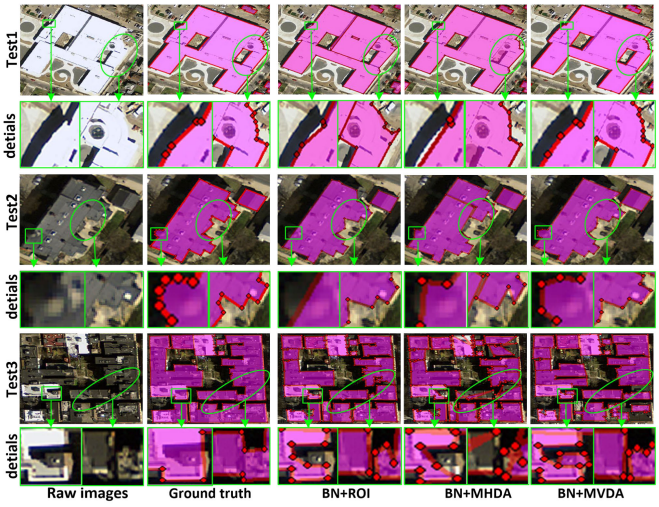


Fig. 8. Different prediction results in the MVDA ablation experiment. The predicted vertices are marked as red dots.

TABLE II
ABLATION ACCURACY RESULTS USING MVDA ON THE DATASETS

Ablation components		Metrics			
		m-IOU	m-AP	mC-IoU	mMTA
BN+	ROI	91.66	80.26	86.23	30.59
	MHDA	91.74	78.45	84.27	32.61
	MVDA	93.90	83.79	88.01	27.73

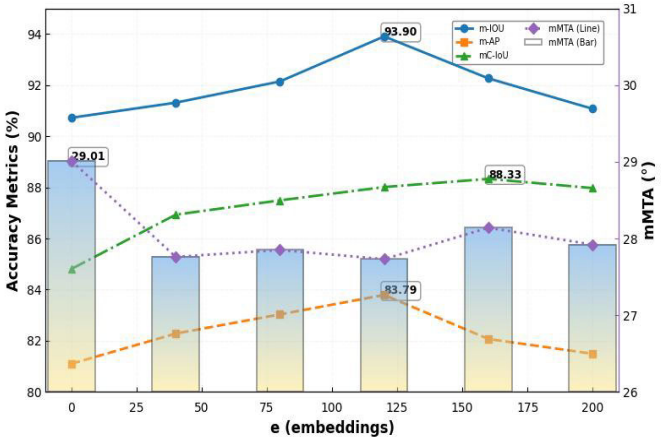


Fig. 9. ECTR module ablation accuracy evaluation.

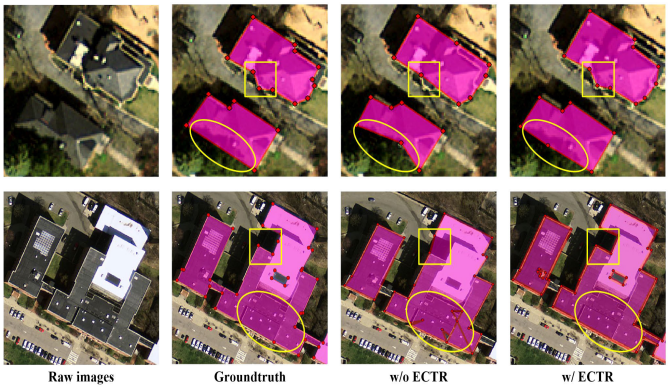


Fig. 10. Visual comparison in ECTR ablation experiments. w/o and w/represent removing and adding a module, respectively.

results and details as shown within the green rectangle and ellipse. Some corners and holes can be correctly identified compared to BN + MHDA. This implies that MVDA can significantly improve vertex prediction accuracy and polygon quality. BM+ROI shows better prediction results than BM + MHDA, but has lower MAT and C-IoU than MVDA. Although MHDA can adaptively match optimal features for each query by designing a deformable offset, building polygons contain many deformable parameters, which significantly increases the burden on the decoders. On the contrary, MVDA and BM + ROI introduce prior query information to focus on the object, thus reducing model optimization difficulty. Additionally, the proposed module combines deformable sampling features and ROI context to enhance the correlation for each instance query group.

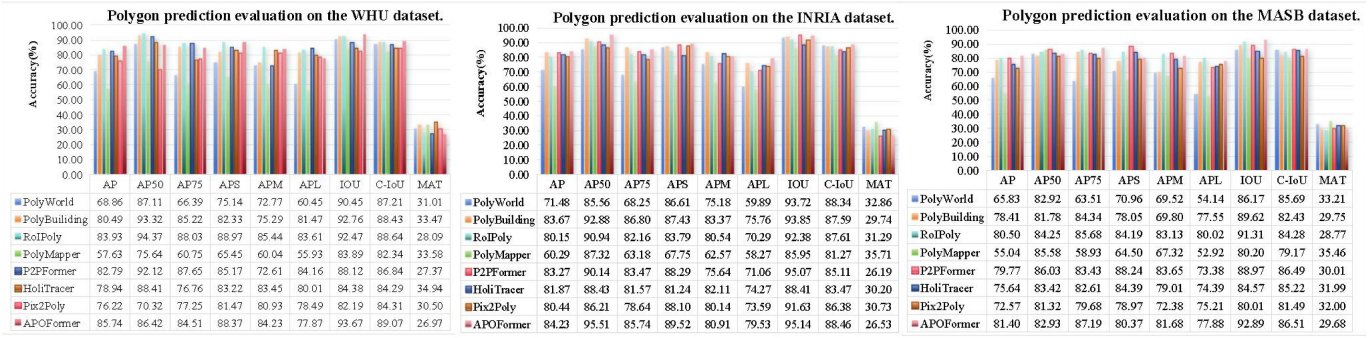
4) Polygonal Topology Reconstruction: In the ECTR module, the parameter e defines the dimensionality of edge-based embeddings. In experiments, the number of polygon vertices N_V was set to 42, and the maximum number of candidate edges N_e was 861, as described in Section II-E. Theoretically, a simple building polygon contains at least N_V edges; thus, e can range from 42 to 861. All candidate vertices interact with these embeddings via attention mechanisms. However, the number of edge embeddings with confidence scores above 0.5 averaged 150 on the three datasets. To balance accuracy and efficiency, the e was restricted to [0, 200] to determine the optimal configuration. Larger e values increase memory consumption. Therefore, the e was linearly varied in increments of 40 to form five control groups. For $e = 0$, the BN relies solely on vertex embeddings as MHSA inputs without the ECTR module.

As illustrated in Fig. 9, the accuracy of building detection increases significantly when e increases from 0 to 120 with an average improvement of 3.18% mIoU and 2.69% mAP. The polygon quality was improved with an average increase of 3.2% mC-IoU and a reduction of 2.28 in mMTA. This indicates that the edge-based context via ECTR enhances decoder prediction performance. However, the accuracy tends to plateau when e reaches 140 or even exhibits a decrease when e>160. Excessive feature edge embeddings increase the computational complexity in MHSA and reduce the inference efficiency of decoders. Therefore, subsequent experiments will use $e = 120$ in ECTR, considering the trade-off between accuracy and efficiency.

Fig. 10 presents the visualization results. ECTR can effectively alleviate the impact of vegetation occlusion on edge connections, as shown in the yellow ellipse of the first row. The backbone model is sensitive to shadow areas where some building corners are missed when vertex features are sampled, as shown in the details of the second row. In addition, there are many incorrect edge connections in highly heterogeneous building areas. In contrast, ECTR reduces these errors by leveraging edge contexts and restoring topological structures.

Under the above experimental configuration, the computational cost of the ECTR module is only 0.085 GFLOP,

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	AP	AP50	AP75	APS	APM	APL	IOU	C-IoU	MAT
PolyWorld	68.86	87.11	66.39	75.14	72.77	60.45	90.45	87.21	31.01
PolyBuilding	80.49	93.32	85.22	82.33	75.29	81.47	92.76	88.43	33.47
RoIPoly	83.93	94.37	88.03	88.97	85.44	83.61	92.47	88.64	28.09
PolyMapper	57.63	75.64	60.75	65.45	60.04	55.93	83.89	82.34	33.58
P2PFormer	82.79	92.12	87.65	85.17	72.61	84.16	88.12	86.84	27.37
HoliTracer	78.94	88.41	76.76	83.22	83.45	80.01	84.38	84.29	34.94
Pix2Poly	76.22	70.32	77.25	81.47	80.93	78.49	82.19	84.31	30.50
APOFormer	85.74	86.42	84.51	88.37	84.23	77.87	93.67	89.07	26.97

	AP	AP50	AP75	APS	APM	APL	IOU	C-IoU	MAT
PolyWorld	71.48	85.56	68.25	86.61	75.18	59.89	93.72	88.34	32.86
PolyBuilding	83.67	92.88	86.80	87.43	83.37	75.76	93.85	87.59	29.74
RoIPoly	80.15	90.94	82.16	83.79	80.54	70.29	92.38	87.61	31.29
PolyMapper	60.29	87.32	63.18	67.75	62.57	58.27	85.95	81.27	35.71
P2PFormer	83.27	90.14	83.47	88.29	75.64	71.06	95.07	85.11	26.19
HoliTracer	81.87	88.43	81.57	81.24	82.11	74.27	88.41	83.47	30.20
Pix2Poly	80.44	86.21	78.64	88.10	80.14	73.59	91.63	86.38	30.73
APOFormer	84.23	95.51	85.74	89.52	80.91	79.53	95.14	88.46	26.53

	AP	AP50	AP75	APS	APM	APL	IOU	C-IoU	MAT
PolyWorld	65.83	82.92	63.51	70.96	69.52	54.14	86.17	85.69	33.21
PolyBuilding	78.41	81.78	84.34	78.05	69.80	77.55	89.62	82.43	29.75
RoIPoly	80.50	84.25	85.68	84.19	83.13	80.02	91.31	84.28	28.77
PolyMapper	55.04	85.58	58.93	64.50	67.32	52.92	80.20	79.17	35.46
P2PFormer	79.77	86.03	83.43	88.24	83.65	73.38	88.97	86.49	30.01
HoliTracer	75.64	83.42	82.61	84.39	79.01	74.39	84.57	85.22	31.99
Pix2Poly	72.57	81.32	79.68	78.97	72.38	75.21	80.01	81.49	32.00
APOFormer	81.40	82.93	87.19	80.37	81.68	77.88	92.89	86.51	29.68

Fig. 11. Performance comparison on building polygon prediction using different methods.

accounting for approximately 0.4% of the ResNet-50 BN with 512 × 512 inputs. This low proportion is mainly due to filtering vertex queries and edges by Top-k confidence scores. This lightweight design achieves an effective balance between accuracy and efficiency, as decoder overhead can be ignored and does not become a bottleneck in the entire pipeline. Therefore, the ECTR module can be easily integrated into frameworks without significant computational losses, while significantly improving the quality of topology prediction.

B. Comparative Experiments Using Different Methods

The proposed method was compared with some representative polygon extraction methods, including PolyWorld, PolyBuilding, RoIPoly, HoliTracer [38], Pix2Poly [39], P2PFormer [31], and PolyMapper [40]. In addition, some instance segmentation algorithms are used to compare pixel-level extraction performance, including SOLOV2, Mask R-CNN, CenterMask, and PolarMask. In the experiment, ResNet50 was used as the backbone encoder to ensure fairness in comparison performance. All configurations remain the same for model training on three public datasets. Three groups of test samples marked as Test 1–Test 3 are selected from WHU, INRIA, and MASB, respectively, for visualization.

As presented in Fig. 11, the proposed method outperforms other models in prediction accuracy and polygonal quality. PolyMapper shows the lowest prediction accuracy compared to other methods on test datasets. This manifests that although it employs ConvLSTM can capture temporal dependencies between vertices, error accumulation causes continuous subsequent vertices offset and distortion of complex building contours. PolyWorld exhibits higher C-IoU, but lower AP compared to other query-based models, especially in the MASB dataset. This model uses attention GNNs to update the building vertex context and predict the adjacency matrix to ensure the correctness of polygon topology relationships. In addition, angle loss can minimize the difference in tangent angle gaps. However, the vertex features from the CNN encoder were not refined and are easily confused with other background information, resulting in an inaccurate prediction of vertex information. Fig. 12 presents some visualization results using different models. Multiple challenges exist in the test dataset to evaluate their predictive performance, including vegetation occlusion, holes, complex shapes, and different scales.

PolyMapper performs well in some regular buildings but cannot handle complex structures such as curved roofs and building holes. In addition, some concave corner areas exhibit offset. PolyWorld presents accurate predictions in some complex building polygons. However, it is more sensitive to vegetation occlusion as marked in the green ellipse. In densely built-up areas, the proposed method has stronger robustness than PolyWorld in distinguishing adjacent buildings. In contrast, PolyBuilding struggles to capture complex edge details and reconstruct the topology structure in easily confused features. Some building corners are missing, and topological errors occur in polygonal corner areas. Similarly, self-intersection error and leak detection exist using the RoIPoly model. Although the proposed method has some topological errors in some building structures, most building polygons can be accurately predicted, especially in areas of occlusion and shadow.

P2PFormer represents primitives as a group of queries sharing a single position embedding. The model effectively attends to multiple endpoints of a primitive using three points for a corner, avoiding the information bottleneck of single-query designs. However, its performance degrades on large-scale buildings, as illustrated in Fig. 10, since the fixed-size ROIAlign grid reduces contextual representation ability. Moreover, P2PFormer is sensitive to occluded areas, where some corner prototypes are incorrectly matched and easily confused with background features, such as details on the INRIA dataset.

In contrast, HoliTracer excels at extracting large-scale building polygons. It employs a multiscale image pyramid and fuses features via a local-global attention mechanism, enabling the model to perceive both fine details and broad surroundings. Nevertheless, its dependence on precise semantic segmentation renders it vulnerable to topological errors when segmentation boundaries are ambiguous. As illustrated in the fourth row of Fig. 12, the results show under-segmentation for the adjacent buildings vectors. Pix2Poly achieves end-to-end gradient flow and eliminates complex post-processing. It exhibits superior performance when predicting mostly regular building polygons, but struggles with complex architectural structures, particularly those featuring highly heterogeneous regions or internal holes. Furthermore, the discrete binning of coordinates introduces quantization errors, causing slight misalignments with query orders.

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Fig. 12. Qualitative comparison with other representative methods for building polygon prediction. The red dots and lines are predicted vertices and edges. The pink polygon was overlaid onto the original image for inspection.

TABLE III

AVERAGE ACCURACY EVALUATION USING DIFFERENT METHODS ON THE DATASETS. BOLDFACE DENOTES THE BEST, AND UNDERLINE DENOTES THE SECOND BEST

Methods	mAP	mAP_{50}	mAP_{75}	mAP_S	mAP_M	mAP_L	mIoU	mC-IoU	mMTA
PolyWorld	68.72	85.2	66.05	77.57	72.49	58.16	90.11	87.08	32.36
PolyBuilding	80.86	89.33	85.45	82.6	76.15	78.26	92.08	86.15	30.99
RoIPoly	81.53	**89.85**	85.29	85.65	**83.04**	77.97	92.05	86.84	29.38
PolyMapper	57.65	82.85	60.95	65.9	63.31	55.71	83.35	80.93	34.92
P2PFormer	81.94	89.43	84.85	**87.23**	77.3	76.2	90.72	86.15	27.86
HoliTracer	78.82	86.75	80.31	82.95	81.52	76.22	85.79	84.33	32.38
Pix2Poly	76.41	79.28	78.52	82.85	77.82	75.76	84.61	84.06	31.08
APOFormer	**83.79**	88.29	**85.81**	86.09	82.27	**78.43**	**93.90**	**88.01**	**27.73**

Table III reports the average accuracy metrics on three datasets. PolyBuilding has higher mIoU and mAP than PolyWorld, but mMTA and mC-IoU are about 3.26% and 2% less than the proposed method. RoIPoly exhibits better performance in medium-scale buildings and mAP_{50}, but the mAP is 2.26% lower than the proposed method. The RoIAlign constraint can avoid background interference and enhance the identification ability for small buildings. However, RoI resolution is rough and cannot capture edge details for large buildings with insufficient geometric information. In contrast, the proposed method integrates edge features and ROI context via CFPE and ECTR modules, significantly enhancing representational capacity to edge-vertex features.

P2PFormer achieves 81.94% mAP with particularly strong performance on small-scale instances, attaining 87.23% mAP_S. However, its performance is less competitive on large buildings and contour regularity, reflected by an IOU of 90.72% and mC-IoU of 86.15%. HoliTracer and Pix2Poly yield moderate results, reaching mAP of 78.82% and 76.41%, both trailing behind P2PFormer in overall precision. In contrast, APOFormer consistently outperforms competing methods, achieving the highest mAP of 83.79%, the largest 93.90% mIoU, and the best 88.01% mC-IoU. Moreover, APOFormer records the lowest 27.73 mMTA, indicating superior contour regularity and holistic extraction quality. These results demonstrate the effectiveness of the proposed modules in capturing fine-grained edge-vertex features and maintaining topological correctness, even for challenging large-scale buildings.

Despite APOFormer's overall superior performance, it exhibits some limitations on individual metrics. Its AP at AP_{50} is comparatively low on the MASB dataset, reaching 86.42%, which trails behind RoIPoly and P2PFormer. This suggests that APOFormer is potentially sensitive to primitive ordering errors. Additionally, APOFormer underperforms on

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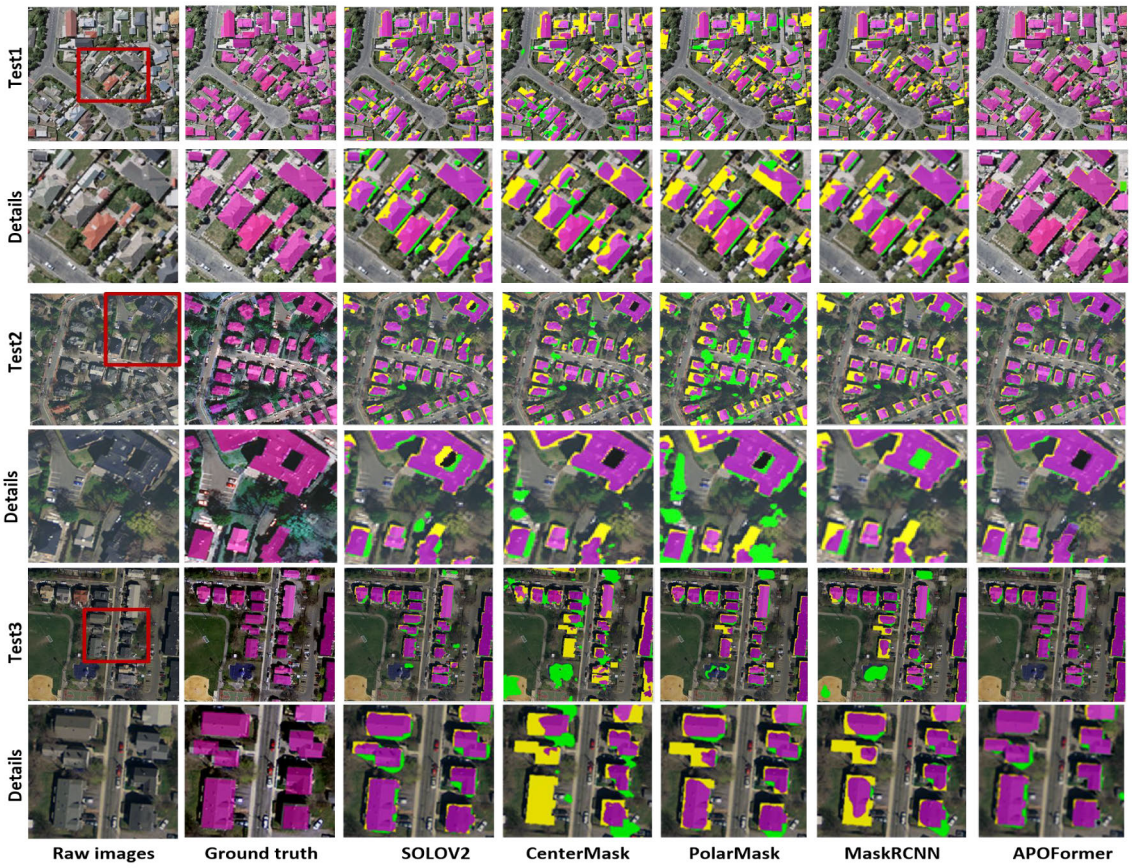


Fig. 13. Qualitative comparison with other SOTA methods for building instance segmentation. The predicted results are overlaid onto the original images, where the purple represents the true positive, the yellow represents false positives, and the green represents incorrect negatives.

TABLE IV
ACCURACY EVALUATION OF INSTANCE SEGMENTATION METHODS

Methods	Datasets WHU		INRIA		MASB		Average metrics	
	IOU	AP	IOU	AP	IOU	AP	mIoU	mAP
SOLOV2	88.37	81.46	92.62	83.01	93.65	72.67	91.55	79.04
MaskRCNN	83.97	65.89	80.64	63.47	85.44	59.83	83.35	63.06
CenterMask	91.28	80.67	85.71	70.09	80.27	68.11	85.75	72.96
PolarMask	85.79	60.94	82.03	61.88	82.16	55.92	83.33	59.58
APOFormer	93.67	85.74	95.14	84.23	92.89	81.40	**93.90**	**83.79**

TABLE V
EFFICIENCY EVALUATION OF METHODS

Models	Parameters (M)	GFLOPs	Inference Time (ms)
APOFormer	48.17	86.74	14.69
P2PFormer	60.39	103.2	18.22
PolyWorld	67.78	135.61	29.84
PolyBuilding	51.60	93.81	17.28
PolyMapper	85.34	154.17	46.15
RoIPoly	42.33	68.50	12.17
HoliTracer	217.34	223.10	83.83
Pix2Poly	**32.14**	**35.69**	**8.27**

large-scale buildings, achieving an AP_L of 77.87% on WHU and 77.88% on MASB, which is lower than RoIPoly with 83.61% and 80.02%. This indicates that the current primitive sampling strategy could be insufficient for capturing fine-grained edge details of large instances, leading to degraded contour completeness for some large buildings.

Fig. 13 presents the instance segmentation results using different methods. Their accuracy evaluation is presented in Table IV. Compared to other mask-based methods, the APOFormer exhibits higher accuracy with 93.90% mIoU and 83.79% mAP. SOLOV2 achieved the second-best segmentation but has many segmentation errors in building boundary areas. In contrast, CenterMask has better performance in large-scale building prediction than SOLOV2. However, its mask prediction exhibits weak performance in small-scale targets. PolarMask shows lower accuracy than other models. This indicates that polar-coordinate-based masks have a rough segmentation accuracy, particularly for building edges. MaskRCNN has a higher mAP than PolarMask, but exhibits lower segmentation accuracy on small-scale buildings. Similarly, its ROIAlign still constrains the feature map scale for fine-grained mask prediction.

This study further completes the evaluation of method efficiency. All efficiency metrics are estimated under a unified experimental setting with an input dimension of $512 \times 512 \times 3$, inference on NVIDIA RTX 4090 GPU. The experiment used 30 images from different scenes to calculate the average efficiency metrics on three public datasets. Finally, algorithm complexity, model size, and inference efficiency are reported in Table V.

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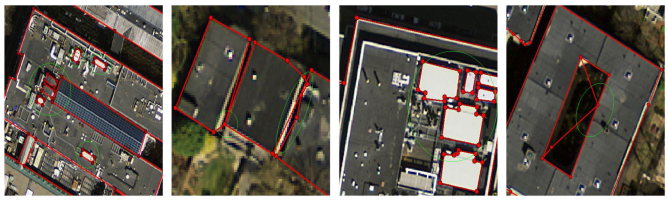


Fig. 14. False cases of building polygon prediction.

Although APOFormer did not achieve optimal performance on metrics, it maintains a well-balanced performance in accuracy and efficiency compared to other approaches. In terms of parameters, RoIPoly and APOFormer are comparatively lightweight (42.33 and 48.17 M, respectively) relative to similar architectures. ROIPoly attains the second-best efficiency by cropping features and significantly reducing query attention interaction within local regions of interest. Pix2Poly exhibits the lightest algorithm complexity and fastest inference speed, but has lower prediction accuracy than APOFormer. PolyWorld constructs global pairing relationships between vertices. Consequently, it incurs longer inference time than APOFormer and higher computational complexity, with 135.61 GFLOPs as the number of vertices increases. PolyBuilding has a similar network architecture to APOFormer, but the complexity increased by 7.07 GFLOPs, and the inference time increased by 2.59 ms due to the lack of global query optimization.

PolyMapper uses VGG-16 as the encoder backbone with a larger parameter quantity. In addition, its RNN processing of longer sequences requires updating multiple bidirectional hidden layers, further increasing computational complexity. P2PFormer establishes corner prototypes to predict polygon sequences, which has a faster inference speed than other methods, yet it remains 3.53 ms slower than APOFormer. This indicates that the decoder spends more time inferring these prototypes due to multiple query representations mapped in attention mechanisms. Notably, HoliTracer has the highest algorithm complexity with 223.1 GFLOPS due to Swin-L as the backbone, increasing the computational burden.

The proposed APOFormer achieves competitive performance through comprehensive quantitative and qualitative evaluations. However, several limitations still require further refinement. As explicitly indicated in Fig. 8, the model exhibits degraded robustness when fitting buildings with smooth curved contours such as circular roofs, mainly because the CFPE module is inherently biased toward extracting sharp corners via Harris responses and regular straight edges. This strong inductive bias inherently restricts the network's representation capability for nonpolygonal structures, especially given insufficient vertex allocation. Meanwhile, the current ECTR module performs favorably in reconstructing outer building boundaries but lacks dedicated topological constraints for inner-ring hole structures. Consequently, obvious topological errors such as self-intersections frequently arise in extremely dense urban regions and buildings with internal courtyards or holes, as marked by the green ellipses in Fig. 14. Additionally, redundant polygon predictions around hole boundaries occasionally appear in large-scale buildings, likely caused by large intra-class feature variations. In future work, this research will incorporate explicit geometric regularization and topological constraints to mitigate these issues. It will also be necessary to explore richer primitive representations and adaptive context optimization strategies for both curved structures and complex topologies.

V. Conclusion

This study proposed a novel method, named APOFormer, for extracting building polygons from remote sensing images. APOFormer employs the CNN-based encoder to generate polygonal vertex queries and utilizes the Transformer structure to construct positional attention and image semantic interaction. Meanwhile, four innovative geometric primitive optimization modules were developed to enhance feature representation and prediction accuracy. Specifically, the learnable primitive detection operator and DQO are designed to generate adaptive vertex embedding. Furthermore, the multi-vertex collaborative attention mechanism and edge-context-guided topology reconstruction effectively improve polygon quality and positional accuracy. APOFormer achieved state-of-the-art performance and demonstrates robust generalization on WHU, INRIA, and MASB datasets. Future research will further improve the accuracy and efficiency of building polygon extraction by exploring multimodal remote sensing data fusion.

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